

# Real World Problem

## Part-1

### Understanding the Business Problem

#### 1. What is the business problem

Beauty Retailers have a large number of skincare and beauty products, which makes it difficult for some customers to find products that match their individual skin concerns. These concerns may include acne, dryness, oiliness, sensitivity, etc. But these customers may not know which products are the most suitable for them.

So, the business could use AI to create a personalized product recommendation system based on the customer's characteristics and already existing reviews. The system would analyze the information about the customers and match it to products that are relevant to customer needs.

This makes it easier for the customers to find products without having to search through thousands of options or even spend time considering and using items which may not even be suitable for them.

#### 2. Why is this problem important to the organization?

Personalized product recommendations can improve the customer's shopping experience by making it quicker and more effective. For the business, the system could increase customer satisfaction, which in turn could potentially increase company sales. The recommendations could also help the company understand which products are preferred by customers with different skin types and concerns.

#### 3. Who is affected by this problem?

Customers, Sales team, Marketing team, Managers, the beauty retailers.

## Part- 2

### Identify the Data

#### 1. What data would you need?

To conduct this project, we would need the following data:

- Customer Skin type
- Customer skin concerns
- Product name
- Product active ingredients
- Product rating
- Product price
- Review text

#### 2. Where could the data come from?

The data could come from the beauty retailer's data catalog and could be customized or partitioned with the help of the already existing customer reviews. The product data could be used to find information like key ingredients, formulations, price and value, and even brand values and packaging. It could also contain customer review information such as review text, skin type, skin tone, ratings, and whether the customer would repurchase the product or not.

#### 3. How much historical data would be useful?

Several years of product and customer data would be necessary. A larger amount of historical data would give the AI system more examples of customers with different characteristics and concerns to

work past. This could help the model identify more reliable patterns, which makes the system more trustworthy.

#### 4. What are the most important variables?

Important variables include:

- Skin Type
- Skin Concerns
- Product ingredients
- Product category
- Product price
- Product rating

#### 5. What could be missing or inaccurate?

Customer information such as skin type or skin tone may vary or be inaccurate depending on each individual and brand; there may be also missing records of it. Some products may have incomplete or outdated ingredient lists. Product categories may be inconsistent, which could make it difficult for the system to match items accordingly. There could also be duplicate product or customer records, which could cause certain products or reviews to have more influence on the model than it should.

## Part-3

### Identify Features and Target.

#### 1. What information would you give the model?

The features could include:

- Customer skin type- oily, dry, combination, sensitive, etc.

- Customer skin concerns- acne, dark spots, redness, etc.
- Customer budget
- Product rating
- Product price

What would you want the model to determine?

- Product name
- Product ingredients
- Product price

For example:

Customer input:

- Skin type: oily
- Skin concern: acne
- Budget: 30\$

System output

- Recommended product: [Product name]
- Key ingredients: [ingredients]
- Price: [price]

## Part- 4

### Select the AI/ML Approach

1. Would you use supervised or unsupervised learning?  
Supervised Learning

2. Explain your choice.

The goal of the system is to take as input a defined set of customer attributes- skin type, skin concerns, budget, and product rating preferences- and produce a specific output- recommended product name, its key ingredients and its price. This is a classic Supervised Learning scenario because it relies on learning a direct mapping pattern between the known inputs and known outputs.

Supervised learning algorithms are trained on labeled datasets, where each data point consists of an input paired with its correct answer. In this model, historical customer data would serve as these labels. By training the model on this data, it learns the patterns connecting customer characteristics to the appropriate product choices which leads it to make accurate predictions for new, unseen customers.

3. Would the problem involve classification or prediction?

Classification

4. Explain your choice.

I would use classification because the goal of the system is to determine whether a product is a good recommendation for a customer. For example: a product could be classified as Recommended or not Recommended based on the customer's skin type, concerns budget and the product information. The model could learn from historical data to identify which types of products were suitable for customers with similar needs. After identifying products that are most likely to be suitable, the system could recommend the best option to the customer.

## Part-5

### Data Quality and Business Impact

#### 1. Identify three possible data-quality problems

- **Missing information:** some customers may not specify their skin concerns, or may have skipped the budget field, or never left a product review or rating. If a significant portion of records have missing values in such key features, the model may learn biased or inaccurate patterns.
- **Inconsistent or Non-Standardized Data:** Attributes like skin type and skin concerns could be particularly vulnerable to this issue, since they're usually entered as free text. For example, 'oily', 'oily skin' and 'oily type' all refer to the same category but could be treated as distinct values by the model. Similarly, prices might be recorded in different currencies or formats.
- **Duplicate records:** The dataset may contain duplicate products or customer reviews, which could give some information more influence than it should have.

#### 2. What could happen if the data is poor?

If the data is poor, the model will tend to learn incorrect patterns between customers and products which could lead to the system recommending a product which might not actually match the customer's concerns.

If the data is biased toward certain segments, the model might also not generalize well to new unseen customer data.

Poor recommendations can directly affect the customer's experience. This could affect the system's reputation and liability.

## Part-6

### Complete AI Solution Summary

Business Problem.

Customers struggle to find skincare or makeup products which are suitable for their unique skin type and concerns, which leads to poor purchases, returns and dissatisfaction.

Data Needed

- Customer skin type
- Customer skin concerns
- Customer budget
- Product name
- Product ingredients
- Product category
- Product price
- Product ratings/reviews
- Customer recommendation information

AI/ML approach

This problem creates an opportunity for an AI skincare/makeup recommendation system that inputs customer characteristics and outputs a matched product.

The system would analyze customer information and historical product data to identify patterns between customer characteristics and products. It would then use these patterns to identify products that are most likely to be suitable for the customer.

### Type of Learning

Supervised Learning- the model can learn from historic labeled examples where the customers have already reviewed, or recommended products.

### Model Output

The main output would be the Product name. The system could also be expected to show the product's ingredients and price.

### Business Benefit

The business could provide customers with a more personalized shopping experience and help them find more effective products quicker. Better recommendations could increase customer satisfaction and trust and potentially lead to increased sales and brand loyalty.

### Potential Risk

Poor quality or biased training data could lead the system to produce inaccurate recommendations, potential skin reactions, customer dissatisfaction and in turn reputational or liability damage to the business.

